

Rebuttal to Reviewer Bkgm

We sincerely thank the reviewer for the careful, constructive, and technically informed assessment of VaSST. We are grateful for the positive recognition of the probabilistic formulation, the conjugate marginalization, the continuous relaxation of symbolic trees, and the practical promise of the method. We have revised the original submission to address the concerns raised, especially regarding experimental breadth, Bayesian model selection, uncertainty quantification, scalability, and positioning relative to VaRT and Bayesian symbolic regression (SR) modules.

1 Modern SR baselines and Experimental Scope

We agree that the original evaluation needed stronger modern symbolic regression baselines. In the revision, we have added comparisons with PySR (Cranmer, 2023), Operon (Burlacu et al., 2020), and DSR (Petersen et al., 2019), in addition to the previously reported BMS, BSR, QLattice, DEAP, and gplearn baselines. Also, we have added experiments on a new Feynman equation $f = \beta(1 + \alpha \cos \theta)$ corresponding to the angular distribution with forward-backward symmetry. These additions directly address the concern that the empirical claims were not sufficiently benchmarked against current state-of-the-art SR systems. We also emphasize that our conclusions are based not only on RMSE but also on the learned symbolic expressions (where we report the symbolic expressions learned from VaSST and competing SR methods), allowing us to assess structural recovery, parsimony, and interpretability. This is especially important (as you have correctly pointed out) in regimes where multiple methods reach the noise floor, since differences at the third or fourth decimal place in RMSE are not scientifically meaningful.

1.1 Results of VaSST and Competing Methods on $f = \beta(1 + \alpha \cos \theta)$

The results for VaSST and all competing methods on the newly added equation $f = \beta(1 + \alpha \cos \theta)$ across different noise levels are reported below.

Method	Learned expression	RMSE
VaSST	$f = -0.8994 + 0.00125 \cos(2\alpha) + 0.10297\theta$ $+ \underline{1.02384 \beta(1 + \alpha \cos \theta)}$	0.13497
PySR	$f = \beta \left[0.009903 \alpha (\theta^2 - 24.691) (7.917 - \theta) + 0.29063 \right] (10.354 - \theta)$	0.45075
Operon	$f = 0.214 - 3.021\beta \left[e^{-0.484\theta} + 0.083\alpha + \frac{e^{-0.500\theta}}{0.080\alpha} \right]$ $- \sin(\sin(0.757\theta)) \{ \sin(0.757\theta) - 0.404\alpha \}$	0.66695
DSR	$f = \beta(1 + \alpha \cos \theta)$	1.848×10^{-15}

Table 1: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
BMS	$\hat{f} = \beta\{1 + \alpha \cos \theta\}$	1.76×10^{-15}
BSR	$f = -0.229931 + 6.94561e^{\cos \theta} + 0.225102\beta^2 - 0.00193193e^\theta$	6.92292
QLattice	$f = 324.462 \left[-0.0214663\alpha + \exp\{-1.90296(0.0865876\alpha - 1)^2 - 4.48049(0.203294\theta - 1)^2\} + 0.0167656 \right] \\ \times \exp\{-2.14832(1 - 0.140906\beta)^2 - 10.6876(0.111068\theta - 1)^2\} + 0.0363653$	0.48031
gplearn	$f = \left[\theta\alpha + \frac{-0.069}{(\beta - 0.593)(0.531 + \beta - \theta)\beta(\beta - \theta) + \beta\theta} \right. \\ + \frac{\theta\alpha + \frac{-0.069}{(\beta - 0.593)(\beta - \theta)} + \frac{\theta}{\theta(\alpha - 0.731)} + (\beta - 0.593)(0.531 + \beta - \theta) + \alpha - \frac{\alpha}{-0.731}}{\theta} + 0.445\theta + \alpha \left. \right] \\ \times \left[\frac{\frac{\beta}{\alpha} + \theta + \beta + (\beta - 0.593)(\beta - \theta)}{(0.531 + \beta)\beta} + 0.445\theta \right] (-0.010\theta) \\ + 0.530\alpha \left[\frac{\theta}{\alpha} + \theta + \beta + (\beta - 0.593)(0.531 + \beta - \theta) \right] + 2\alpha$	6.31679
DEAP	$f = \frac{\alpha^2}{\theta + \left(\frac{\beta}{\theta} - \beta\right)} = \frac{\alpha^2}{\theta + \frac{\beta}{\theta} - \beta}$	13.54156

Table 2: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ in the noiseless setting from BMS, BSR, QLattice, gplearn, and DEAP along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$f = -0.92928 + 0.000575 \cos(2\alpha) + 0.10661\theta \\ + 1.02407 \beta(1 + \alpha \cos \theta)$	0.16513
PySR	$f = (-2.48391\theta + 20.96369 + \alpha) \\ \times \beta(\theta - 4.86540)(-0.24211\theta + 1.95075)$	1.49573
Operon	$f = 0.377 - 83.187 \left[\frac{\cos(-0.981\theta)(-0.368\beta)}{\sqrt{1 + (-1.085\theta)^2}} \right. \\ \times \left. \frac{1}{\sqrt{1 + \left(\frac{\cos(\cos(\cos(-0.999\theta)))}{\sqrt{1 + \exp\{\cos(-0.999\theta) \cos(0.340\alpha)\}^2}} \right)^2}} \right] (0.318\alpha)$	0.51157
DSR	$f = \beta(1 + \alpha \cos \theta)$	0.09908

Table 3: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
BMS	$f = \beta(1 + \alpha \cos \theta)$	0.10022
BSR	$f = -0.232683 + 6.94597e^{\cos \theta} + 0.225131\beta^2 - 0.00193157e^\theta$	6.92420
QLattice	$f = -0.619856\alpha + 19.747(3.03165 - 0.377702\theta)(1.28896\beta + 0.0343982)$ $\times \exp\{-12.6002(1 - 0.129159\theta)^2 - 2.02944(0.0893216\alpha - 1)^2\} + 3.45018$	0.53195
gplearn	$f = [\theta\alpha + \{(-0.097 + \beta)(\alpha - \beta) - (\theta + \beta) - \theta\}] \left[\left(\frac{0.649}{\theta} + 0.445\theta \right) (-0.010\theta) \right]$ $+ \beta(\beta - 0.593)(\beta - \theta) \left[(-0.097 + \beta) \left\{ \frac{\alpha - \beta}{\theta} - \frac{-0.069}{(0.531 + \beta) \left[\frac{-0.069}{\beta + \frac{\beta\{0.833 + \beta - [\theta - (\theta - \beta)]\}}{\beta\theta}} \right] + 0.247} \right\} + \theta \right]$ $+ 2\alpha$	6.36929
DEAP	$f = \frac{\alpha^2}{\theta + \left(\frac{\beta}{\theta} - \beta\right)} = \frac{\alpha^2}{\theta + \frac{\beta}{\theta} - \beta}$	13.54231

Table 4: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise level $\sigma = 0.1$ from BMS, BSR, QLattice, gplearn, and DEAP along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$f = -0.95916 - 0.000097 \cos(2\alpha) + 0.11026\theta$ $+ \frac{1.02430 \beta(1 + \alpha \cos \theta)}{\theta}$	0.23748
PySR	$f = \left[-0.02377 \left\{ \theta + -0.70132(\theta - 9.13492)^2 \right\}^2 + 1.22422 \right] \alpha\beta$	1.32648
Operon	$f = 3.429 + 0.352 \left[\left\{ \frac{\cos(0.976\theta)}{\sqrt{1 + (0.653\theta)^2}} \right\} (5.549\beta)(2.578\alpha) \right.$ $\left. + \left\{ \frac{\cos(-0.781\theta)}{\exp(\cos(1.718\theta))} \right\} (-0.437\alpha) - 1.536\alpha \right]$	0.46428
DSR	$f = \beta(1 + \alpha \cos \theta)$	0.1981

Table 5: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise $\sigma = 0.2$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
BMS	$f = \beta(1 + \alpha \cos \theta)$	0.20045
BSR	$f = -0.235435 + 6.94633e^{\cos \theta} + 0.225159\beta^2 - 0.00193121e^\theta$	6.92692
QLattice	$f = -68.9593(0.00212903 - 0.280842\beta)$ $\times \left[-0.0561226\alpha + \exp\{-2.07305(1 - 0.0903278\alpha)^2\} \right.$ $\left. - 3.37702(-0.109637\theta - 1)^2(0.157901\theta - 1)^2\right] + 0.0526521\right] + 0.0674459$	0.50443
gplearn	$f = \{\theta\alpha + (\theta - \beta)\} \left[\left(\frac{0.649}{\theta} + 0.445\theta \right) (-0.010\theta) \right]$ $+ \beta(\beta - 0.593)(\beta - \theta) \left[(-0.097 + \beta)A + \theta + B \right] + 2\alpha$	6.42362
DEAP	$f = \frac{\alpha^2}{\theta + \left(\frac{\beta}{\theta} - \beta \right)}$	13.54380

Table 6: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise level $\sigma = 0.2$ from BMS, BSR, QLattice, gplearn, and DEAP, along with their RMSEs. For compactness, the gplearn expression uses the shorthands $A = \frac{\alpha - \beta}{\theta} - \frac{-0.069}{\beta\theta}$, and $B = 0.833 + \beta - [\theta - (\theta - \beta)] + \frac{\theta\beta + \theta/\beta}{\theta}$, where $C = \left[(-0.097 + \beta) \left(\frac{\alpha - \beta}{\theta} - \frac{-0.069}{\frac{-0.097 + \beta}{\frac{0.649}{\theta} + 0.445\theta}} \right) + \theta + \beta \right] (\beta - 0.593)(\beta - \theta)/\beta\theta + \frac{-0.069}{(\beta - \theta)(\beta + 0.247) + \frac{-0.097 + \beta}{D}}$ with $D = \left[\left(\frac{-0.069}{\beta + \frac{\beta(0.833 + \beta - [\theta - (\theta - \beta)])}{\beta\theta}} \right) + 0.247 \right] (\beta + 0.247)(\alpha - \beta)$.

1.2 Results of VaSST, PySR, Operon, and DSR for Representative Feynman Equations

The results for VaSST and newly added state-of-the-art SR benchmarks (PySR, Operon, DSR) for representative Feynman equations $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ and $P = \kappa A(T_2 - T_1)/d$ across different noise regimes.

Method	Learned expression	RMSE
VaSST	$\Delta U = -0.000714 + 0.000914 \frac{G^2 m_1^3 m_2}{r_1^2}$ $-0.996541 \frac{Gm_1 m_2}{r_1} + 1.010681 \frac{Gm_1 m_2}{r_2}$	0.003656
PySR	$\Delta U = \frac{Gm_1 m_2}{r_2} - \frac{Gm_1 m_2}{r_1}$	8.61×10^{-16}
Operon	$\Delta U = -0.998001 \frac{Gm_1 m_2}{r_1} + 1.000232 \frac{Gm_1 m_2}{r_2}$	1.34×10^{-5}
DSR	$\Delta U = \frac{Gm_1 m_2}{r_2} - \frac{Gm_1 m_2}{r_1}$	8.70×10^{-16}

Table 7: Learned symbolic expressions for Feynman equation $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$\Delta U = -0.003874 + 0.000609 \frac{G^2 m_1^3 m_2}{r_1^2} - 0.995991 \frac{G m_1 m_2}{r_1} + 1.017123 \frac{G m_1 m_2}{r_2}$	0.100194
PySR	$\Delta U = \frac{G m_1 m_2}{r_2} - \frac{G m_1 m_2}{r_1}$	0.100224
Operon	$\Delta U = -0.998605 \frac{G m_1 m_2}{r_1} + 4.985798 \frac{\frac{G m_1 m_2}{r_1}}{\sqrt{1 + \left(4.921 \frac{r_2}{r_1}\right)^2}}$	0.100157
DSR	$\Delta U = \frac{G m_1 m_2}{r_2} - \frac{G m_1 m_2}{r_1}$	0.09908

Table 8: Learned symbolic expressions for Feynman equation $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$P = 0.014782 - 0.805371 \left[\frac{AT_1 \kappa}{d} \right] + 0.998520 \left[\frac{A \kappa T_2}{d} \right] - 0.000254 \left[\frac{1}{\sqrt{A} T_1 \kappa} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{A} T_1 \kappa - \frac{T_2}{T_1}} \right]$	0.012602
PySR	$P = \frac{A \kappa (T_2 - T_1)}{d}$	1.59×10^{-16}
Operon	$P = 4.668 \frac{-2.018665 \left(\frac{T_2}{T_1}\right) (\sqrt{A} T_1 \kappa) + 2.051 (\sqrt{A} T_1 \kappa)}{1.904 \sqrt{A} T_1 \kappa} - 9.425 \frac{d}{\sqrt{A}} \sqrt{1 + \left[\frac{-4.07664 \left(\frac{T_2}{T_1}\right) (\sqrt{A} T_1 \kappa)}{\sqrt{1 + \left\{ \sin\left(-0.172 \frac{T_2}{T_1}\right) + 0.516 \frac{d}{\sqrt{A}} \right\}^2}} \right]^2}$	9.66×10^{-5}
DSR	$P = \frac{A \kappa (T_2 - T_1)}{d}$	1.61×10^{-16}

Table 9: Learned symbolic expressions for Feynman equation $P = \kappa A(T_2 - T_1)/d$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$P = 0.019505 - 0.842846 \left[\frac{AT_1\kappa}{d} \right] + 1.005597 \left[\frac{A\kappa T_2}{d} \right]$ $- 0.000149 \left[\frac{1}{\sqrt{AT_1\kappa}} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{AT_1\kappa} - \frac{T_2}{T_1}} \right]$	0.100325
PySR	$P = \frac{A\kappa(T_2 - 1.0040724T_1)}{d}$	0.100212
Operon	$P = 0.538 - 0.678 \left[\frac{\exp\left(-0.525\frac{T_2}{T_1}\right) + 1.888\sqrt{AT_1\kappa} - 1.856922\left(\sqrt{AT_1\kappa}\right)\left(\frac{T_2}{T_1}\right)}{\sqrt{1 + \left(1.254\frac{d}{\sqrt{A}}\right)^2}} \right]$ $+ \sin\left(\sin\left(\exp\left[\exp\left\{-1.164401\left(\sqrt{AT_1\kappa}\right)\left(\frac{T_2}{T_1}\right)\right\}\right]\right)\right)\right]$	0.099843
DSR	$P = \frac{A\kappa(T_2 - T_1)}{d}$	0.09908

Table 10: Learned symbolic expressions for Feynman equation $P = \kappa A(T_2 - T_1)/d$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

2 Model-class Limitation and Internal Constants

We agree with you that the present VaSST formulation does not fully support arbitrary internal constants inside symbolic compositions. In the current implementation, constants enter primarily through the outer Bayesian linear regression layer (as regression coefficients), while leaves are selected from input features. We now explicitly acknowledge this modeling scope in the revision. At the same time, some expressions with apparent internal constants can still be represented through tree composition; for example, $\sin(2x)$ can be expressed as $\sin(x + x)$. Nevertheless, expressions requiring genuinely free internal constants, such as $\sin(cx)$ with unknown non-integer c , are outside the current model class. We now state this clearly as a limitation and identify learnable internal constants as a natural extension of VaSST for future work.

3 Posterior-aware Symbolic Expression Selection

We agree that, given the Bayesian structure of VaSST, posterior information can be more directly leveraged when summarizing and ranking the learned symbolic expressions. Accordingly, in the revision, we introduce a posterior-aware selection criterion based on the *joint marginal posterior distribution of the symbolic tree structures* given the data viz.,

$$\log \text{JMP}(\mathcal{T}_1, \dots, \mathcal{T}_K) = \log \pi(\mathcal{T}_1, \dots, \mathcal{T}_K \mid \mathcal{D}_n) = \log p(y \mid \mathbf{X}, \mathcal{T}_1, \dots, \mathcal{T}_K) + \log \pi(\mathcal{T}_1, \dots, \mathcal{T}_K),$$

up to additive constants independent of tree structures. Here, $p(y \mid \mathbf{X}, \mathcal{T}_1, \dots, \mathcal{T}_K)$ denotes the collapsed marginal likelihood obtained after analytically integrating out the regression coefficients and the noise variance under the conjugate Normal-Inverse-Gamma (NIG) prior, while $\pi(\mathcal{T}_1, \dots, \mathcal{T}_K)$ denotes the structural prior over the symbolic trees. This quantity admits a closed analytical form (to be derived in the revised Appendix).

After learning the variational distribution over tree structures, we sample hard symbolic trees and rank the resulting candidate symbolic forests using $\log \text{JMP}(\mathcal{T}_1, \dots, \mathcal{T}_K)$. Thus, RMSE remains an important predictive diagnostic, but the primary symbolic summary is now tied directly to posterior support. This revision makes fuller use of the Bayesian machinery in VaSST: the method quantifies uncertainty not

only over regression coefficients, but also over competing symbolic explanations. In particular, when the signal is strong, the posterior may concentrate around a parsimonious symbolic structure, whereas in more ambiguous regimes it may spread across several structurally distinct but similarly predictive expressions. We now report these posterior-aware top-ranked expressions, making the uncertainty contribution more explicit and better aligned with UAI's emphasis on principled uncertainty-aware inference.

Results corresponding to the newly added Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noiseless and noise levels $\sigma = (0.1, 0.2)$ are reported below, where rank the top symbolic expressions learned from VaSST according to the maximum value of $\log \text{JMP}(\mathcal{T}_1, \dots, \mathcal{T}_K)$. Results of selection based on maximum $\log \text{JMP}(\mathcal{T}_1, \dots, \mathcal{T}_K)$ for all other Feynman equations will feature in the revised submission. For each case, VaSST is executed with $K = 3, D = 3$ and other hyperparameters are configured to be the same as in Appendix H.1 of the original submission.

Rank	Posterior/JMP-selected learned expression	RMSE	log JMP
1	$f = 0.001817 + 0.000007 \alpha \theta^2 \cos \beta - 0.000963 \alpha + \underline{1.000653 \beta(1 + \alpha \cos \theta)}$	0.100246	1486.529922
2	$f = -0.005438 - 0.002204 \cos\{\cos(\beta) + \alpha \theta\} + \underline{1.001241 \beta(1 + \alpha \cos \theta)} - 0.013960 \cos \theta$	0.100160	1482.196908
3	$f = -0.746272 + \underline{1.001444 \alpha \beta \cos \theta} + \underline{0.966006 \beta} + 0.966006 \cos(\cos \alpha) + 0.000151 \beta \theta^3$	0.184680	394.598913
4	$f = 3.438256 - 0.062553\{\cos(\cos \alpha) + \theta + \alpha \theta\} + \underline{0.964345 \alpha \beta \cos \theta} + 0.166902(\cos \theta + \alpha \beta)$	0.402627	-1139.109250
5	$f = 0.855936 - 0.001083\{\cos \beta + \alpha \beta^2 \theta\} + 0.000508 \alpha^2 \cos \beta \cos(\alpha \beta) + \underline{0.997444\{\beta + \alpha \beta \cos \theta\}} + 0.997444 \cos \beta$	0.491626	-1532.211112

Table 11: Top five VaSST posterior samples ranked by log JMP for the Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise level $\sigma = 0.1$.

Rank	Posterior/JMP-selected learned expression	RMSE	log JMP
1	$f = -0.004888 - 0.001243 \theta \cos \alpha + \underline{1.000162 \alpha \beta \cos \theta} + \underline{1.001182 \beta}$	0.200301	263.370061
2	$f = 0.116199 + 0.125104\{\beta \theta + \cos(\beta \theta)\} - 0.017039 \cos(\beta \cos \alpha) + \underline{1.034378 \alpha \beta \cos \theta}$	0.320475	-653.345352
3	$f = 3.151643 - 0.586528 \alpha + 0.023327(\alpha \beta \theta + \theta) + \underline{1.034875 \alpha \beta \cos \theta}$	0.378674	-996.677919
4	$f = 0.966897 + \underline{0.994034 \alpha \beta \cos \theta} + \underline{0.712135 \beta} + 0.712135 \cos(\beta + \theta) + 0.433517 \cos \theta$	0.452181	-1327.977398
5	$f = 1.054896 + 0.690847 \cos(\beta + \theta) + \underline{1.012862 \alpha \beta \cos \theta} + \underline{0.690847 \beta} + 0.002719 \cos\{2\theta(\beta + \theta)\}$	0.467602	-1396.007323

Table 12: Top five VaSST posterior samples ranked by log JMP for the Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise level $\sigma = 0.2$.

Rank	Posterior/JMP-selected learned expression	RMSE	log JMP
1	$f = 0.000679 - 0.000051 \theta \cos \alpha + \underline{1.000244 \alpha \beta \cos \theta + 0.999361 \beta}$	0.002333	2932.122750
2	$f = 0.125702 + 0.124846\{\beta\theta + \cos(\beta\theta)\} - 0.014547 \cos(\beta \cos \alpha) + \underline{1.034366 \alpha \beta \cos \theta}$	0.253871	-198.584675
3	$f = 3.138430 - 0.582044 \alpha + 0.023266(\alpha\beta\theta + \theta) + \underline{1.034832 \alpha \beta \cos \theta}$	0.324166	-703.220380
4	$f = 0.970430 + \underline{0.994147 \alpha \beta \cos \theta + 0.711760 \beta} + 0.711760 \cos(\beta + \theta) + 0.431951 \cos \theta$	0.400733	-1090.290750
5	$f = 1.057883 + 0.690591 \cos(\beta + \theta) + \underline{0.690591 \beta + 1.012901 \alpha \beta \cos \theta} - 0.007236 \cos\{2\theta(\beta + \theta)\}.$	0.417907	-1174.613731

Table 13: Top five VaSST posterior samples ranked by log JMP for the Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under the noiseless regime.

4 Novelty of VaSST and Differences with VaRT and other Bayesian SR methods

We have sharpened the discussion of VaSST relative to VaRT (Salazar, 2023), Bayesian Symbolic Regression (BSR, Jin et al., 2020), and the Bayesian Machine Scientist (BMS, Guimerá et al., 2020).

While both methods (VaSST and VaRT) use variational ideas over tree structures, the objects and inference problems are fundamentally different. VaRT targets variational inference for decision/regression trees with data-partitioning semantics, whereas VaSST performs symbolic regression over compositional mathematical expression trees. This requires several nontrivial innovations:

1. **Symbolic operator semantics:** internal nodes represent mathematical operators such as $+$, $-$, $\times$, $/$, $\sin$, $\cos$, $\log$, $\exp$,² rather than decision splits. The tree therefore defines an analytical symbolic expression, not a partition of the covariate space.
2. **Soft symbolic evaluation:** VaSST introduces a differentiable symbolic evaluation mechanism that propagates mixtures over operator and feature choices through the expression tree. This is substantially different from soft routing in decision trees.
3. **Gumbel-Softmax/Binary-Concrete relaxation for symbolic structures:** VaSST uses reparameterized relaxations for operator, feature, and expansion decisions in symbolic expression trees. These relaxations are adapted to the compositional SR setting, where invalid or numerically unstable symbolic operations must be handled carefully.
4. **Collapsed Bayesian linear layer:** VaSST places a conjugate NIG prior on the outer regression coefficients and analytically marginalizes them, yielding a collapsed structural objective. This is not present in VaRT and is central to the Bayesian symbolic regression formulation.
5. **Posterior over symbolic expressions:** The posterior object in VaSST is a distribution over symbolic expressions/forests, enabling uncertainty quantification over scientific laws. This is distinct from uncertainty over predictive tree partitions.

Also, VaSST is distinct from MCMC-based Bayesian symbolic regression methods such as BSR and BMS. The key distinctions are as follows:

1. **Inference mechanism:** BMS and BSR explore the symbolic expression space using Markov chain Monte Carlo over discrete tree structures. While this gives a principled Bayesian formulation, the resulting search is inherently combinatorial and can mix slowly across structurally distant symbolic modes. In contrast, VaSST relaxes the discrete symbolic tree posterior into a differentiable variational family using Binary-Concrete and Gumbel-Softmax reparameterizations, enabling gradient-based optimization over symbolic structures.
2. **Prior specification:** BMS uses a prior constructed from corpus-based regularities over mathematical expressions. VaSST instead uses an explicit depth-dependent symbolic tree prior that directly encodes parsimony by discouraging unnecessarily deep symbolic trees.
3. **Optimization over symbolic structures:** BSR relies on discrete proposal mechanisms and local MCMC moves over expression space. VaSST performs continuous optimization over relaxed operator, feature, and expansion probabilities, allowing structural learning to be carried out with stochastic-gradient methods.
4. **Collapsed Bayesian objective:** VaSST places a conjugate Normal-Inverse-Gamma prior on the outer regression coefficients and analytically marginalizes the regression coefficients and noise variance. This yields a collapsed marginal likelihood for each symbolic forest, which is used both during training and in posterior-aware symbolic selection through the joint marginal posterior $\log \text{JMP}(\mathcal{T}_1, \dots, \mathcal{T}_K)$.

Therefore, the novelty of VaSST is not merely that it places a prior over symbolic expressions. Rather, VaSST combines a coherent Bayesian symbolic forest prior, differentiable soft symbolic evaluation, reparameterized variational inference over operators/features/expansion indicators, and closed-form conjugate marginalization of regression coefficients and noise variance. This produces a collapsed, gradient-optimizable Bayesian symbolic regression objective, while still allowing posterior sampling of hard symbolic expressions after training. We now explicitly highlight these distinctions in the revised literature review so that VaSST is positioned not as a direct transfer of VaRT or a minor variant of existing Bayesian symbolic regression methods, but as a new probabilistic symbolic regression framework for uncertainty-aware discovery of scientific expressions.

5 Discussion on the Relaxed Surrogate and the Exact Discrete ELBO

We thank you for raising this important point. We agree that the surrogate ELBO $\widehat{\mathcal{L}}(\phi)$ in Eq. (14) should be interpreted more carefully, and in the revision we will explicitly distinguish the exact ELBO of the original discrete symbolic-tree model viz., $\mathcal{L}(\phi)$ in Eq. (9) from the relaxed objective optimized in VaSST viz., $\widehat{\mathcal{L}}(\phi)$.

For the original discrete model, the ELBO is

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi(\Theta)} [\log p(y \mid T)] - \text{KL}\{q_\phi(\Theta) \parallel \Pi(\Theta)\},$$

where T is the hard symbolic design matrix induced by discrete expansion, operator, and feature assignments. In contrast, VaSST optimizes the temperature-controlled relaxed objective

$$\mathcal{L}_\tau(\phi) = \mathbb{E}_{P_{\phi, \tau}^{\text{soft}}} [\log p(y \mid T_{\text{soft}})] - \text{KL}\{q_\phi(\Theta) \parallel \Pi(\Theta)\},$$

where $P_{\phi, \tau}^{\text{soft}}$ is induced by the Binary-Concrete and Gumbel-Softmax relaxations, and T_{soft} is the resulting soft symbolic design matrix. Therefore, Eq. (14),

$$\widehat{\mathcal{L}}(\phi) = \widehat{\mathcal{L}}_{\tau, S}(\phi) = \frac{1}{S} \sum_{s=1}^S \log p(y \mid T_{\text{soft}}^{(s)}) - \text{KL}\{q_\phi(\Theta) \parallel \Pi(\Theta)\},$$

is an unbiased Monte Carlo estimator of $\mathcal{L}_\tau(\phi)$, not of the exact discrete ELBO $\mathcal{L}(\phi)$. We will state this explicitly in the revision.

This also clarifies the approximation gap:

$$\mathcal{L}(\phi) - \hat{\mathcal{L}}_{\tau,S}(\phi) = \underbrace{\mathcal{L}(\phi) - \mathcal{L}_\tau(\phi)}_{\text{relaxation bias}} + \underbrace{\mathcal{L}_\tau(\phi) - \hat{\mathcal{L}}_{\tau,S}(\phi)}_{\text{Monte Carlo error}}.$$

The second term is standard stochastic approximation error. The first term is the relaxation bias caused by replacing hard symbolic trees with continuous soft trees. This bias is expected because the collapsed marginal likelihood is nonlinear in the symbolic design matrix: after integrating out (β, σ^2) , $\log p(y | T)$ depends on T through $T^\top T$, Σ_n , $|\Sigma_n|$, and b_n . Hence, in general,

$$\mathbb{E}_{q_\phi}[\log p(y | T)] \neq \mathbb{E}_{P_{\phi,\tau}^{\text{soft}}}[\log p(y | T_{\text{soft}})].$$

We will revise the paper to make clear that VaSST optimizes the ELBO of a relaxed symbolic tree model viz., $\mathcal{L}_\tau(\phi)$, rather than the exact ELBO of the original discrete model viz., $\mathcal{L}(\phi)$.

The justification for this surrogate is algorithmic and inferential. Direct optimization of the discrete ELBO would require either score-function estimators over symbolic structures or explicit combinatorial search over an exponentially large tree space, both of which are high-variance or computationally prohibitive. The relaxed objective instead provides a differentiable reparameterized surrogate with low-variance gradients. As the temperature is annealed, the Binary-Concrete and Gumbel-Softmax variables become increasingly concentrated near Bernoulli and categorical draws, so the learned soft structures progressively approach hard symbolic trees. Thus, the relaxation is not presented as exact at finite temperature, but as a controlled optimization device for learning variational parameters over symbolic structures.

Importantly, inference and reporting ultimately return to the original discrete symbolic tree space. After optimization, VaSST samples hard expansion indicators, operators, and features from the learned Bernoulli and categorical variational factors and applies the deterministic pruning map to obtain valid symbolic trees. In the revision, these hard symbolic forests are then ranked using the posterior-aware joint marginal posterior distribution

$$\log \text{JMP}(\mathcal{T}_1, \dots, \mathcal{T}_K) = \log p(y | \mathbf{X}, \mathcal{T}_1, \dots, \mathcal{T}_K) + \log \pi(\mathcal{T}_1, \dots, \mathcal{T}_K),$$

up to constants independent of the tree structures. Therefore, the continuous relaxation is used to make posterior learning tractable, while uncertainty quantification and symbolic model selection are carried out over sampled hard symbolic expressions.

We will revise accordingly: Eq. (14) will be described as a Monte Carlo estimator of a temperature-controlled relaxed ELBO, the relaxation bias relative to the exact discrete ELBO will be explicitly acknowledged, and the role of the relaxation will be framed as a principled computational approximation for posterior learning over symbolic expressions.

6 Ablation Study w.r.t. K and D

Here we give a detailed overview of the sensitivity of VaSST w.r.t. number of trees K and depth D . We conducted an ablation study analyzing the impact of: (i) number of trees K and (ii) tree depth D . For each of the Feynman equations considered, VaSST is implemented for different values of

$$K \in \{2, 3, 4, 5\}, \quad D \in \{2, 3, 4, 5\},$$

keeping all other hyperparameters fixed (as in Appendix H.1 of the original submission). These new results will feature in the Appendix, along with a discussion of VaSST’s sensitivity with respect to K and D .

A highlight of the ablation study for two representative Feynman equations

$$P = \frac{\kappa A(T_2 - T_1)}{d}, \quad \Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

is as follows. Gaussian noise with standard deviation 0.1 has been added to the data. See Tables 14—17. The key findings are:

- **Performance stabilizes with model complexity.** For both equations, RMSE improves significantly when moving from $K = 2$ to $K \geq 3$, and stabilizes for $K \geq 4, D \geq 3$. This indicates that VaSST does not require large model sizes to achieve optimal predictive accuracy.
- **Low-capacity regimes can limit symbolic recovery.** As expected, insufficient model capacity leads to occasional failures in recovering the true expression. For

$$P = \frac{\kappa A(T_2 - T_1)}{d},$$

failure occurs at $(K = 2, D = 5)$, and for

$$\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

failures occur at $(K = 2, D \in \{2, 4, 5\})$ and $(K = 3, D = 5)$. This highlights that both sufficient K and appropriate D are necessary for capturing compositional structure.

- **Runtime scales predictably with K and D .** Runtime increases approximately linearly in K and exponentially with D , reflecting the growth in the number of nodes. Importantly, moderate configurations (e.g., $K = 3, D = 3$) already achieve near-optimal RMSE, offering a favorable accuracy–efficiency trade-off.
- **Principled heuristics for choosing K and D .** Based on the ablation results, we include practical guidelines for selecting K and D . Since VaSST is designed to recover simple and interpretable scientific expressions, our recommendation follows an Occam’s razor principle in-built in the depth dependent split probability prior: choose the smallest model capacity that attains stable RMSE and symbolic recovery. Empirically, $K \in \{3, 4\}$ and $D \in \{3, 4\}$ provide a favorable accuracy–interpretability–runtime trade-off, while larger depths increase runtime exponentially and may lead to unnecessarily complex symbolic expressions without substantial predictive gains.

Table 14: Runtime for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	50.6	97.2	191.3	401.6
3	62.1	112.0	221.8	498.5
4	72.2	125.0	249.8	628.2
5	83.1	146.0	299.4	803.4

Table 15: RMSE for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.1019	0.1002	0.1002	0.1070
3	0.1000	0.1011	0.1012	0.1019
4	0.1000	0.0999	0.1000	0.1000
5	0.0999	0.0999	0.0998	0.1007

Table 16: Runtime for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	54.7	98.8	192.4	399.6
3	61.5	112.0	221.4	491.3
4	67.9	123.1	251.8	600.9
5	72.3	134.8	286.4	800.5

Table 17: RMSE for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.2330	0.1012	0.1959	0.2349
3	0.1002	0.1019	0.1051	0.1918
4	0.1168	0.1002	0.1005	0.1012
5	0.1002	0.1005	0.1004	0.1010

7 Symbolic Equivalence and Duplicate Hard Trees

We agree that symbolic equivalence is a fundamental issue inherent to the symbolic regression problem itself. Different trees may represent identical or nearly equivalent mathematical functions, such as algebraically rearranged forms such as $\log(x_1 \times x_2) = \log x_1 + \log x_2$ or $x_1 + x_2 = x_2 + x_1$. In the current VaSST framework, the prior and posterior are defined over symbolic tree expressions, not quotient classes of mathematical functions under algebraic equivalence. Therefore, when sampling hard trees, equivalent structural variants may appear. We now explicitly acknowledge this point. Our current goal is to recover parsimonious symbolic expressions that fit the data and reveal interpretable structure, rather than to perform complete equivalence-class canonicalization. We identify posterior inference over equivalence classes of symbolic expressions as an important direction for future work.

8 Response to Minor Comments

1. We have clarified the use of protected operations such as `safe_div`, `safe_inv`, etc., both in the anonymized GitHub README file and in the revised Appendix.
2. We also clarify that the released artifact includes a lightweight toy configuration for quick reproducibility checks, whereas the experimental configurations reported in the Appendix of the original submission are the actual settings used to generate the main results.
3. Now, we have also added the unary operators “exp” and “log” to the released code (Anonymous Github).
4. In addition to the previous and newly added competing methods, we have also added a discussion on the following references in the revised literature review: Reinforcement Symbolic Regression Machine (Xu et al., 2024) and Call for Action: towards the next generation of symbolic regression benchmark (Guilherme et al., 2025).
5. We agree that budget equalization is important, and we have revised the experimental discussion to make the comparison protocol more explicit. Since VaSST, BMS/BSR, GP-based methods (PySR, Operon, gplearn, DEAP), and neural SR methods (QLattice, DSR) use different computational primitives, exact equality of “iterations” is not directly meaningful: one VaSST gradient step, one

MCMC transition, one GP generation, and one neural-policy update correspond to very different amounts of symbolic search. Therefore, our comparison controls the shared ingredients that can be matched across methods—train/test splits, operator sets, symbolic grammar where supported, and evaluation metrics. Appendix H in the original submission already documents the method-specific configurations, including VaSST’s 2000 optimization steps and $H = 2000$ hard-tree posterior samples, BMS/BSR’s 2000 MCMC iterations, and the GP-based/QLattice configurations. In the revision, we further add state-of-the-art baselines and present compute-aware comparisons through predictive accuracy, structural recovery, parsimony, posterior support, and computational cost. This provides a fairer and more informative comparison across heterogeneous SR algorithms.

6. Finally, we corrected the Denison et al., 1998 citation.